

Visual ERP-based brain-computer interface use with severe physical, speech and eye movement impairments: case studies

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September 15, 2025

Abstract

Background: Individuals who experience severe speech and physical impairment face significant challenges in communication and daily interaction. Visual brain-computer interfaces offer a potential assistive solution, but their usability is limited when facing restrictions to eye motor control, gaze redirection and fixation eye motor control. This study investigates the feasibility of a gaze-independent visual oddball brain-computer interface for use as an augmentative and alternative communication device in the presence of limited eye motor control. *Methods:* Seven participants with varying degrees of eye motor control were recruited and their conditions were thoroughly assessed. Visual oddball brain-computer interface decoding accuracy was evaluated under three conditions: overt, covert, and free visuospatial attention using multiple decoding methods. *Results:* covert visuospatial attention with central fixation leads to decreased accuracy, whereas free visuospatial attention is comparable to overt visuospatial attention in some participants. Furthermore, cross-condition decoder training and evaluation suggests that training with overt visuospatial attention may

improve performance in brain-computer interface users experiencing gaze control difficulties. *Conclusions:* These findings highlight the need for adaptive decoding strategies and further validation in applied settings in the presence of gaze impairment.

Keywords: brain-computer interface, augmentative and alternative communication (AAC), event-related potentials, covert visuospatial attention, eye motor impairment

1 Background

Neurological conditions, such as acquired brain lesions, neuromuscular diseases and Amyotrophic Lateral Sclerosis can result in speech and physical impairment. This, in turn, significantly alters how individuals participate in daily activities and communication, sometimes leading to a reduced quality of life. Augmentative and alternative communication technology [Ascari2018, Elsahar2019, Curtis2022] leveraging visual brain-computer interfaces [Schultz2017, Peters2022], which relies on the interpretation of visual stimuli by the user, offers several advantages in this context. The rapid stimulation pace of visual brain-computer interfaces and the modulation of information over spatial attention allow for high information transfer rates [Abiri2019, Han2023]. Combined with their ability to operate with non-invasive recording technology, this makes them well-suited for communication tasks.

However, visual and eye motor differences such as nystagmus (uncontrolled eye movements), diplopia (double vision), ophthalmoplegia (partial or complete eye paralysis), fatigability and a limited range of head motion can significantly interfere with accurate control eye gaze, reducing performance with brain-computer interfaces that rely on visual cues [McCane2014, FriedOken2020, Pasqualotto2015, VanDenKerchove2024a]. Unfortunately, it is again for this group that eye tracking solutions also perform poorly, making them more reliant on potential developments in brain-computer interfaces that do not rely on eye gaze.

Hence, some brain-computer interface and augmentative and alternative communication solutions rely on the capability of the user to direct their gaze at a desired target. In this case, the gaze fixation of the user and their visuospatial attention coincide (overt visuospatial attention). With the development of brain-computer interface-augmentative and alternative communication for users with eye movement impairment in mind, alternative solutions usually circumvent gaze redirection by designing gaze-independent brain-computer interfaces [Riccio2012], which either avoid visual stimulation or exploit some form of covert visuospatial attention, where gaze and visuospatial attention do not coincide [Aloise2012a, Zhang2010, Thiery2016, Pires2011, Egan2017].

Several studies on visual oddball brain-computer interfaces indicate that performance declines when

relying on covert visuospatial attention [Brunner2010, Treder2010, RonAngevin2019]. The tasks performed in these studies emulate covert visuospatial attention using central gaze fixation. This implicitly assumes that brain-computer interface users with eye motor impairment who would benefit from gaze-independence would feel comfortable performing this central fixation. In other words, one could argue that a brain-computer interface verified to work only with central fixation could also be considered gaze-dependent. It does not account for the remaining eye motor capabilities of individuals experiencing speech and physical impairment and most forms of eye movement difficulties, the (dis)comfort experienced when fixating, and other challenges and confounding factors related to eye motility.

Several of the aforementioned works improved gaze-independent brain-computer interfaces in various ways, or discovered obstacles preventing accurate gaze-independent brain-computer interface use. However, it is notable that most studies reporting on gaze-independent visual brain-computer interfaces were performed with participants without neuromuscular conditions, and only very few report on brain-computer interface use of participants with speech, physical and gaze impairment. Results are usually different from those obtained with control participants without neuromuscular conditions, due to differences in capabilities, brain response, experimental equipment and environment.

Combaz2013 showed that steady-state visually evoked potential and oddball paradigm brain-computer interfaces can be used effectively by individuals living with Locked-in Syndrome. **Lesenfants2014** tested a brain-computer interface using gaze-independent steady-state visually evoked potentials in six participants with Locked-in Syndrome yet only two of them exceeded chance level accuracy. More recently, **Peters2020** performed a trial with two participants living with a more progressed stage of Amyotrophic Lateral Sclerosis with visual impairment. Their steady-state visually evoked potential paradigm was not optimized for gaze-independence, but the system showed high accuracy, outperforming an eye tracking alternative. It would be of interest to determine whether such results can be replicated with participants with other neurological conditions, and with a visual oddball rather than an steady-state visually evoked potential paradigm.

Orhan2012 and **Oken2014** tested the rapid serial visual presentation speller with participants with Locked-in Syndrome. Due to the serial nature of the stimulation paradigm, communication was rather slow but they demonstrated the feasibility of the rapid serial visual presentation paradigm in a relevant target population.

Severens2014 evaluated the visual Hex-o-Spell [Treder2010] on 5 participants living with Amyotrophic Lateral Sclerosis and showed that this visual oddball interface optimized for gaze-independence can outperform a tactile brain-computer interface. While this speaks to the power of visual paradigms even in groups that are expected to have eye motor impairment, they did not verify the gaze direction of participants during the experiment. It was assumed that participants were performing overtly. Participants living with

Amyotrophic Lateral Sclerosis also exhibited a substantially lower accuracy than healthy controls (58% vs. 88%).

Previously mentioned approaches focus mainly on optimizing the stimulation paradigm and visual layout. However, an alternative approach was proposed by **Aloise2012**. They retained classical paradigms with proven effectiveness but adapted decoding strategies to enhance decoding performance of covert visuospatial attention. While the attempts of **Aloise2012** did not discover any classifiers improving covert decoding, **Arico2014** made a step in this direction by relying on properties of the covert visuospatial attention event-related potential. **Arico2014** identified that latency jitter in the P3 event-related potential component is significantly higher when performing covert visuospatial attention compared to overt visuospatial attention for multiple event-related potential stimulation paradigms. Hence, they proposed an adapted decoding scheme which aligns the relevant event-related potential components over trials. While they show the potential effectiveness of this approach, they stopped short of building an applicable decoder. **Hardiansyah2020** also leveraged the distribution of event-related potential latencies to decode covert visuospatial attention.

VanDenKerchove2024 built on this latency jitter property to implement an applicable decoder to improve gaze-independent decoding for the visual Hex-o-Spell interface. They propose a classifier using temporal alignment to improve decoding performance in the presence of jitter. Furthermore, this study also partially accounted for the fact that brain-computer interface users with speech, physical and gaze impairment might not fully rely on central gaze fixation by evaluating settings independent of central fixation. They showed gaze-independent performance can be improved in healthy subjects by using a suited decoding strategy that corrects for latency jitter in covert visuospatial attention responses. Yet, there is a need for verification of these results in the presence of speech, physical and gaze impairment.

Ultimately, this line of research aims to develop gaze-independent brain-computer interface for individuals with complete Locked-in Syndrome and have no alternative means of communication. However, recruitment of these potential brain-computer interface users can be challenging, since they are very few and may have extensive accessibility needs. Furthermore, they may experience challenges in performing the experiment task when participating in a study [**Wolpaw2006**]. On the other hand, individuals experiencing only partial loss of voluntary eye motor control or living with a less progressed stage of a neuromuscular disease, can also experience difficulties operating classical augmentative and alternative communication technology based on eye-tracking. Hence, there is a need to verify if earlier methods and concepts actually apply for them. They could also benefit from solutions tailored to their specific situation and remaining capabilities. Finally, we also notice that literature often follows a strict dichotomy between overt and covert visuospatial attention with cued fixation. It is not obvious that such cases occur in natural visual brain-computer interface by individuals with speech, physical and gaze impairment.

Based on literature and our experience, we formulate the following research questions. (1) Do brain-computer interface users with speech, physical and gaze impairment naturally rely on covert or overt visuospatial attention, or a mix of both? (2) If they rely on covert visuospatial attention, how well do they perform using a visual brain-computer interface? (3) Do they also benefit from decoding approaches improving covert visuospatial attention decoding? (4) If the user can perform a mix of covert and overt visuospatial attention, can cued overt or covert visuospatial attention be exploited to craft a suited calibration and operation strategy to increase performance and comfort? To this end, we apply the concepts from earlier work and literature, especially the WCBLE decoder and Hex-o-Spell visual oddball interface the presence of speech and physical impairment and various differences in eye motor control. The contributions of this study are as follows: (1) the recruitment of individuals that specifically have speech, physical and gaze impairment in a visual brain-computer interface study and the exploration of their eye motility and experienced comfort when operating such a brain-computer interface, (2) a baseline evaluation of the performance of a gaze-independent visual brain-computer interface for this group, (3) the verification of the gaze-independent decoding strategy proposed by **VanDenKerchove2024**, (4) and a calibration strategy suitable for these users. Finally, we formulate a set of recommendations based on our experiences when carrying out this study, which could be of interest when designing further similar studies or brain-computer interface-augmentative and alternative communication solutions for this target user population.

2 Methods

2.1 Recruitment and study design

Participants were recruited from the Neuromuscular Reference center at University Hospitals Leuven (Leuven, Belgium), TRAINM Clinics (Antwerp, Belgium), the Neurorehabilitation Unit at the Lille University Medical Center (Lille, France), and Fondation Partage et Vie (Loos, France). Experiments were performed after prior informed consent and under the supervision of the treating physician, with a protocol prior approved by the Ethics Commission of the University Hospitals Leuven (S62547).

We partially base the recruitment criteria on the brain-computer interface user classification system presented by **Wolpaw2006**. They present a three-class system: (1) complete Locked-in Syndrome without voluntary movements (including eye movements), (2) the ability to control a single-button switch using very limited voluntary movement, consisting of eye movements or small limb movements, and (3) reduced but substantial voluntary movements (either speech or limb control), able to operate a range of classical augmentative and alternative communication devices.

In order to be included in the study, participants must:

1. be at least 18 years old and no older than 60 years,
2. belong to class 2 or 3 according to the aforementioned brain-computer interface user classes,
3. have limitations to the extent or comfort of their eye motor control (partial or full gaze paralysis, uncontrolled gaze movements, or conditions of the capability to direct the gaze or fixate),
4. have given their informed consent prior to participation.

Participants were excluded if:

1. they had any diagnosis of a major medical condition other than neuromuscular conditions covered by inclusion criteria 2 and 3,
2. they had a predisposition to or a history of any kind of epileptic seizures, including photosensitive epilepsy,
3. they had a severe loss in vision or hearing that would significantly impair participation in the experiment,
4. they were using specific psychoactive medications or substances that could affect the outcome (neuroleptics or benzodiazepines),
5. the instructions or the experimental tasks could not be made accessible for participation
6. they experienced any other limitations preventing them from performing the given task.

The study was conducted in two visits. If two visits were feasible, the first visit served to verify the inclusion and exclusion criteria, to explain the context and course of the study, and answer the participant's questions in cooperation with the paramedical team. Afterwards, the informed consent form was signed. Initial clinical and eye motor assessment was also performed in the first visit. The first visit was conducted by the clinician or paramedical team, or by the experimenting researchers together with the clinician or paramedical team. If a participant was included, the second visit consisted of further clinical, eye motor and visual assessment, the electroencephalography setup, eye tracker setup and brain-computer interface stimulation and recording. The second visit lasted around two to three hours.

Clinical assessment was based on a questionnaire polling for basic demographic and neuromuscular questions, answered by the participant or the paramedical team. This questionnaire is provided in ???. Other relevant details concerning communication, pragmatics and presentation of the patient were observed by the

researchers who carried out the experiments or through conversation with the participant or the paramedical team. The clinical assessment was complemented with further relevant details provided by the supervising clinician.

We also asked the participants to evaluate their mood and tiredness on a scale from 1 to 10 before and after the experiment conducted in the second session. This should serve as a preliminary indication of comfort and task load, since it was impractical to conduct more extensive surveys.

2.2 Vision and eye motor examination

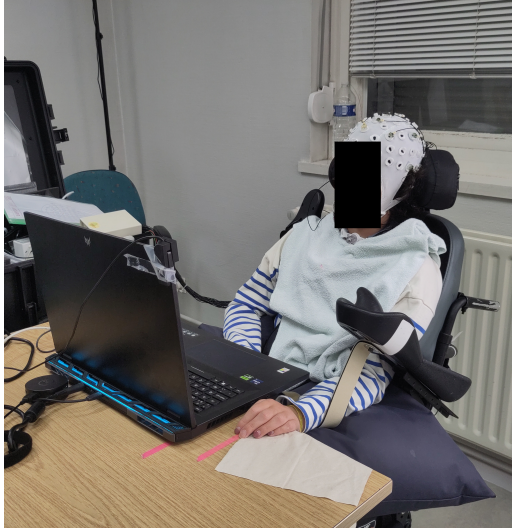
Eye motor and visual differences assessments were obtained through self-reporting using the questions presented in ?? and completed with details from the medical file or from an oculomotor assessment provided by the clinician where possible.

Vision and eye motor control were assessed using the ‘visual skills’ framework established by **FriedOken2020**. These describe on a higher level the factors that can reduce visual brain-computer interface performance when impaired. Visual acuity was assessed using a logMAR chart [**Bailey1976**]. During this assessment and during the rest of the experimentation session, participants wore their corrective glasses or lenses where applicable. In table 2, the distinction between minor and major impairment was made according to the following rule: Major impairment implies the impairment was complete, or the given ability on its own prevents or nearly prevents the participant from operating a regular visual oddball BCI as a healthy control subject would, after potential corrections. Minor impairment implies the ability is limited to discomfort or does not on its own strictly prevent the user from operating a visual BCI as a healthy control subject would, after potential corrections. Finally, we also recorded gaze position throughout the experimental session to register the participant’s gaze relative to the stimulated brain-computer interface targets, as described below in section 2.4.

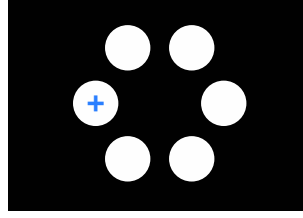
2.3 BCI stimulation

The brain-computer interface stimulation procedure was based on the Hex-o-Spell [**Treder2010**] implementation presented by **VanDenKerchove2024**. During the experimental session, participants were comfortably seated in chair or in their wheelchair in front of a table. Stimuli were presented on an Acer Predator Helios laptop with an 18” screen (Acer, Inc., Taiwan) placed at a 60 cm distance. Stimulation was performed using PsychoPy (version 2023.1.3) [**Peirce2019**]. A Cedrus StimTracker (Cedrus Corp., CA, USA) ensured synchronization of stimuli with the recorded electroencephalography.

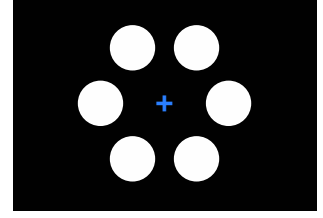
The stimulation display shows 6 circular white targets with a diameter of 4.15° visual angle laid out on



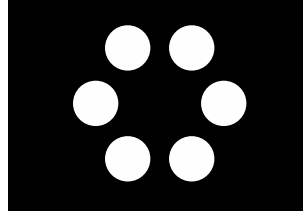
(a) A participant seated in a wheelchair in front of the stimulation laptop with EEG cap.



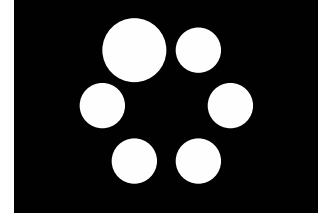
(b) Stimulation interface with 6 targets and fixation crosshair positioned for *overt* visuospatial attention.



(c) In *covert* visuospatial attention, the fixation crosshair is placed in the center of the screen.



(d) In *free* visuospatial attention, no fixation crosshair is displayed.



(e) Targets are intensified by enlarging them for 100 ms.

Figure 1: Stimulation and recording setup for the oddball brain-computer interface experiment with different visuospatial attention conditions

the corners of a hexagon with radius 12.28° . The task consists of counting the intensifications of a cued target, while fixating the gaze on a cued fixation cross. If the fixation cross was absent, the gaze could move freely during the counting task. Targets were intensified by scaling them to a larger size (5.50° of visual angle)

Using this system of an attention cue (the target to count), and a gaze cue (the fixation cross), three different visuospatial attention settings were explored as shown in fig. 1. In the overt visuospatial attention setting, the participant was instructed to fixate on the cued target or to try this to the maximum extent of their capability or comfort. In the covert visuospatial attention setting, the participant was instructed to fixate on the center of the screen, to the extent of their ability. In addition to **VanDenKerchove2024**, an additional *free visuospatial attention* setting was introduced. Here, the participant was instructed to perform the task as they deemed most comfortable. This allowed us to investigate the user's natural way of operating the brain-computer interface given their individual differences in vision and eye motor control. Participants were instructed to avoid head movements if they could do so voluntarily.

Stimulation was performed in blocks, one block for each cued target for each visuospatial attention condition, resulting in $6 * 3 = 18$ blocks presented in randomized order.

At the beginning of each block, the attention cue (the target to count) is indicated by a yellow cue. Next, the yellow arrow disappeared and, the gaze fixation cross was shown if applicable. This cross stayed visible throughout the block. The participant was asked to convey a simple signal if they were ready to start

a block, and if they had memorized the attention cue and fixation cue, on which the experimenter could progress through the experiment at the pace of the participant.

After the cue, all 6 targets began intensifying in pseudorandom order. A target was intensified for a duration of 100 ms with an inter-stimulus interval of 200 ms with added random jitter uniformly distributed between -50 ms and 50 ms. Within a block, the intensification count of a target was randomly varied between 10 and 15 intensifications to keep the relevance of the counting attention task. Only the first 10 intensifications of each target were kept for analysis. After the stimulation phase of the block, the participant was asked for the number of intensifications counted to verify if they were performing the task correctly.

The experiment started with a practice block for each of the visuospatial attention conditions with instructions for each condition. During these practice blocks, the participant received oral feedback on their counting accuracy to ensure they understood and were able to perform the task. After block 6 and 12, the participant took a break with a minimum of 5 minutes.

2.4 Data collection & preprocessing

During the experimental session, electroencephalography and eye tracking signals were recorded. Electroencephalography was recorded at 1000 Hz using the Neuroscan Neuvo portable amplifier (Compumedics Neuroscan, Australia) connected to a second laptop for registration. The electroencephalography headset used 18 active AgCl electrodes (EASYCAP GmbH, Germany) placed on a cap according to the international 10-20 layout. Using electrolyte gel, electrode impedances were reduced below 10 k Ω . Additionally, the electrooculogram was recorded.

The electroencephalography was band-pass filtered between 0.5 and 16 Hz. Bad channels were rejected using the RANSAC algorithm [Fischler1981] and visual inspection. Next, the electroencephalography was re-referenced to the average of mastoid electrodes TP9 and TP10, and independent component analysis was performed to reject artifactual components based on correlation with the electrooculogram or by visual inspection. For decoding, epochs were cut from -0.1 to 0.9 s relative to stimulus onset, and no baseline correction was performed. ?? presents the event-related potentials obtained as the contrasts of the average target and non-target trials for each subject and visuospatial attention setting. These event-related potentials shown for visual interpretation originate from baseline-corrected epochs cut from -0.2 to 1.0 seconds relative to stimulus onset.

Eye tracking was performed using a Tobii X2-30 Compact (Tobii Technology AB, Sweden) portable eye tracker placed at the bottom of the laptop screen. The registered data was cleaned by fusing left and right eye screen-based gaze coordinates into one channel for the horizontal and one for vertical gaze position. If

both were present for a given sample, the fused channel was the mean of both values. If at a given sample either the left or the right eye was not detected for a given channel, the value of the other one was adopted. If both were missing, the gaze position remained unset at that time point, and no interpolation was performed.

2.5 Data analysis

2.5.1 Range of comfortable eye motility

Research question 1 aims to explore the eye motor capabilities and usage during overt, covert and free visual brain-computer interface use. We clarify this by evaluating the eye gaze position of the participant with target gaze fixation (overt visuospatial attention) central fixation (central visuospatial attention) and to assess the occurrence of these patterns when gaze is not cued (free visuospatial attention). After calibration, eye tracking follows the participant’s gaze throughout the brain-computer interface recording session when stimulation is active. The outcome of this analysis should be interpreted with care, as eye tracker calibration partially relies on intact eye motility. The participant’s position relative to the eye tracker might also frequently shift throughout the experimental session despite our best efforts, due to e.g. a regular need for aspiration of the tracheostoma of some participants.

2.5.2 Brain-computer interface decoding

Research question 2 aims to establish a baseline brain-computer interface target selection decoding performance for the target user group. To this end, we perform off-line decoding of the cued target selections using the experimental protocol described in section 2.3 and two state-of-the art decoding algorithms: block-Toeplitz linear discriminant analysis [Sosulski2022] and the Riemannian approach XDAWN-Cov+TS+LDA [Cecotti2017]. For decoding, cropping the epoch between 0 ms and 800 ms, which resulted in maximal accuracy.

Target selection accuracies are obtained using 6-fold cross-validation with folds corresponding to stimulation blocks. All target stimulus repetitions beyond 10 are dropped. We report both single-trial accuracy and accuracy averaged over specific numbers of repetitions of all 6 targets. In the single-trial case, each subsequent set of 6 unique trials are considered as a selection, allowing for more selections than 1 per block resulting in fine-grained accuracies. In the case of averaging over multiple trials, only one selection per block is made to allow for comparison over multiple repetitions as in realistic operation. The target and non-target epochs are separately averaged over these repetitions to enhance their signal-to-noise ratio.

Accuracies are reported with 95% confidence intervals, determined using 10,000 bootstrapping repetitions. To interpret the data, these confidence intervals were used to test hypotheses verifying which conditions or

decoders were outperformed by others.

2.5.3 Gaze-independent brain-computer interface decoding

Research question 3 investigates the role of gaze-independent decoding algorithms could play in improving brain-computer interface decoding performance for individuals with speech, physical and gaze impairment. We repeat the decoding analysis above using a decoding algorithm specifically optimized for covert visuospatial attention decoding: Classifier-based Latency Estimation with Woody iterations [VanDenKerchove2024]. Classifier-based Latency Estimation with Woody iterations is specifically designed for gaze-independent decoding by relying on latency jitter correction.

Classifier-based Latency Estimation with Woody iterations integrates a regular classifier with temporal alignment in an iterative scheme. First, the data is convolved with the trained regular classifier, in this case block-Toeplitz linear discriminant analysis. From this convolution, classification score over time is extracted, and the latency with maximum classification score can be selected. These latencies can then be used to align the data to retrain the classifier for another iteration.

To allow for proper latency estimation, a wider region of interest from -100 ms to 900 ms relative to stimulus onset was used.

2.5.4 Cross-condition decoding

Finally, research question 4 seeks an alternative approach to improving decoding performance. We propose the following approach: performance or comfort might be improved if calibration is performed using a given visuospatial attention setting, executed by relying maximally on the participants remaining gaze control. This way, informative properties depending on gaze fixation can be included in the decoder parameters, independent of gaze position during operation. This in turn might improve performance when multiple different visuospatial attention conditions are performed during realistic brain-computer interface operation.

Off-line target selection accuracy was assessed as a metric of performance in a way similar to the brain-computer interface decoding analysis. However, the cross-validation scheme was altered to train the decoder on 5 blocks from a given visuospatial attention condition and evaluated on a block from another given visuospatial attention setting. This way, all pairs of overt, covert and free visuospatial attention are evaluated.

ID	Diagnosis	Age	Sex	Speech	Primary communication method	W	KB
A1	bulbar-onset ALS	58	M	anarthric	tablet with touchscreen	3	4
B1	FRDA	41	M	dysarthric	speech	3	3
B2	FRDA	43	F	dysarthric	speech	3	3
B4	FRDA	48	M	dysarthric	speech	3	3
C2	ischemic brainstem stroke	43	M	anarthric	eye movements	2	4
C3	hemorrhagic brainstem stroke	43	F	anarthric	letterboard with finger pointing or touch	2	3
C4	ischemic brainstem stroke	54	M	anarthric	letterboard with finger pointing or touch	2	3

Table 1: Included participants with their diagnosis and relevant communication capabilities. W: classification according to **Wolpaw2006** used as inclusion criterion, KB: classification according to **Kuebler2008** for further reference.

	A1	B1	B2	B4	C2	C3	C4
Visual fixation	—	—	—	—	—	—	—
Eyelid function	+	+	+	+	+	—	—
Ocular motility	+	—	+	—	/	/	—
Binocular vision	+	+	+	+	—	/	/
Field of vision	+	+	+	+	+	/	—
Involuntary movement	+	/	/	—	—	/	+
Visual acuity	0.0	0.0	0.6	0.2	0.0	0.0	0.6

Table 2: Vision and eye motor assessment as defined by **FriedOken2020** of participants included in this study. + no impairment, — minor impairment, / major impairment. Visual acuity was assessed using the logMAR scale (0.0: normal vision, 0.05-0.5: minor visual impairment, 0.5-1.3: major visual impairment, > 1.3 blindness [**WHO2019**]).

3 Results

3.1 Included subjects and their visual capabilities

In total, 11 individuals were contacted. Of these, one person with Multiple Sclerosis was excluded based on criterion 3. A person recovering from traumatic brain injury was excluded based on criteria 2 and 4, and one person recovering from stroke based on criterion 1. One further person recovering from a stroke was excluded due to technical difficulties during the experimental session. Ultimately, 7 participants were retained. Of these, one participant was diagnosed with bulbar-onset Amyotrophic Lateral Sclerosis, three with Friedreich’s Ataxia and three were recovering from Locked-in Syndrome due to stroke. Table 1 lists the included participants and their diagnoses, each case is clinically described in ???. For these participants, the second visit of the study was carried out either in the hospital (A1), in their rehabilitation clinic (B1, B2, B4) or in their care home (C2, C3, C4).

Eye motor and visual differences assessments are reported in table 2. All participants reported they experienced some degree of fatigue or discomfort when fixating. Participant A1 had the mildest impairment, only reporting fatigue when fixating for prolonged times. The participants with Friedreich’s Ataxia mostly had eye tremors and impaired pursuit. Impairment of eye motor function of was most prominent in participants C2, C3, and C4, those recovering from brain stem and cerebellar stroke.

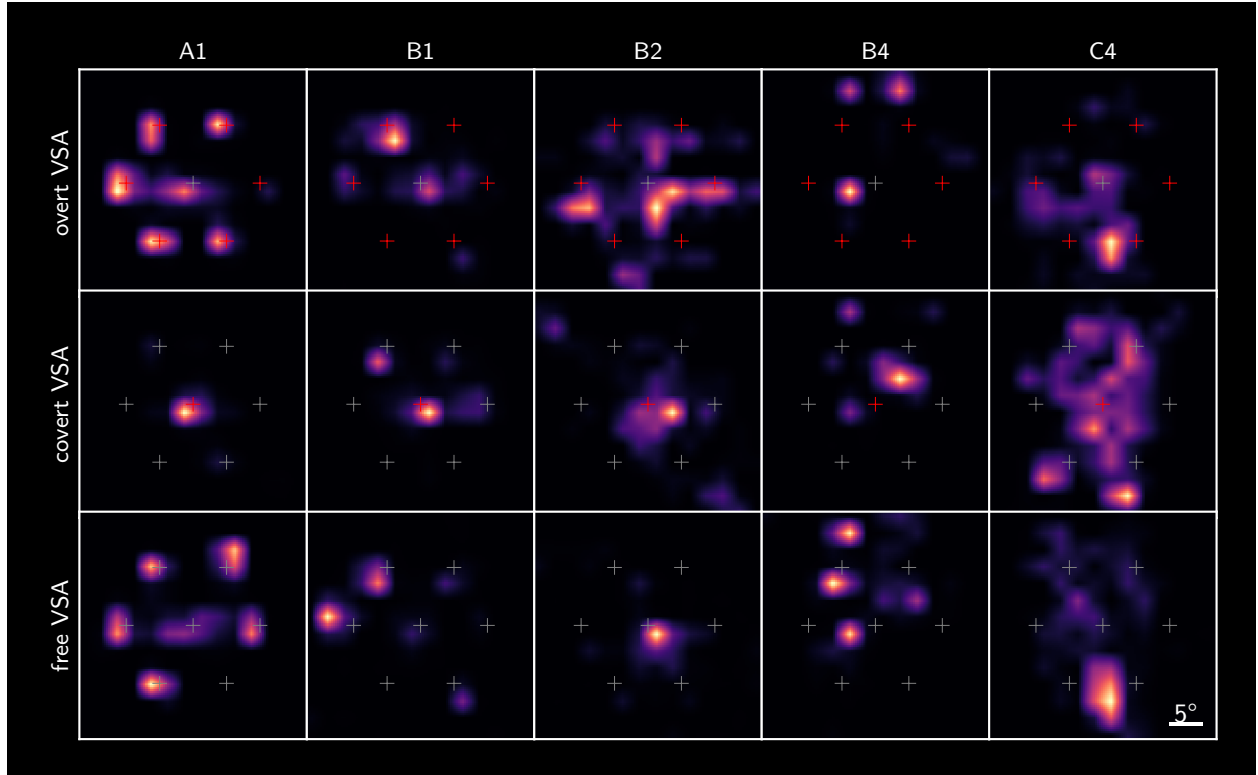


Figure 2: Distribution of the recorded gaze position during the experimental session in the three visuospatial attention conditions. Crosshairs represent stimulus positions, with positions cued during the given condition indicated in red. Eye tracking calibration could not be successfully completed for participants C2 and C4. Subjects B2 and C4 preferred covert brain-computer interface operation, with B2 resting gaze near the middle of the screen, and C4 near the bottom. A high-resolution, non-downsampled figure is available as Additional file 1.

Figure 2 maps gaze position relative to the targets across conditions recorded during the brain-computer interface recording session. A1 had relatively intact gaze control and was able to correctly perform the cued overt and covert settings. When gaze was uncued, he fixated on the cued target. This was also mostly the case for B1, although eye tracking revealed that he chose not to perform central gaze fixation when cued in at least one of the stimulation blocks. We were unable to record his gaze near the bottom-left stimulus position, either due to eye tracker failure or because the participant was not comfortable fixating on this position. Eye tracker calibration did not succeed for subject B4, but given the transformation of gaze positions to the stimulus space, he was assumed to be overtly performing the free task.

B2 was able to perform overt visuospatial attention and central fixation to some extent, yet eye tracking shows a larger spread in gaze position compared to A1 and B1. In the free visuospatial attention condition, however, she preferred to attend stimuli covertly when the gaze was uncued. This was confirmed by the participant.

The overt and central gaze fixation settings were also not properly adapted to participant C4. In the free visuospatial attention condition, eye tracker results show that his gaze was usually near the bottom

Subject	A1	B1	B2	B4	C2	C3	C4
Mood before	6	6	7	9	10	10	10
Mood after	7	7	7	5	10	-	10
Tiredness before	6	5	7	6.5	3	10	9
Tiredness after	7	5	7	6.5	3	-	9

Table 3: Participant experience of the visual brain-computer interface task.

two targets, indicating some degree of covert or split visuospatial attention. Technical difficulties were encountered while recording gaze position with the Tobii X2-30 Compact for participants C2 and C3, since they both had one eye that was occluded respectively by the prism glass and the eyelid. Both participants reported they could not fixate on some of the stimuli.

As a complementary metric to the participants’ ability to perform the task and use similar brain-computer interface systems we questioned their experience of the experiment. Mood and tiredness were generally comparable before and after the task. Only participant reported a decrease from 9 to 5. Contrary to our expectations, the experiment did not have a large impact on tiredness. Only participant A1 reported an increase from 6 to 7.

3.2 BCI decoding performance

Figure 3 shows cross-validated single-trial target selection accuracy for the evaluated visuospatial attention settings for the different decoders. A full table of results is available as Additional file 2. Accuracy results do not reveal a clear trend in decoder performance per condition. In the covert visuospatial attention setting with cued central gaze fixation, accuracy was lower overall. Target selection accuracy for this task was around chance level (1/6) for participants B4, C2 and C3. Classifier-based Latency Estimation with Woody iterations did not improve over the accuracy of block-Toeplitz linear discriminant analysis in the free visuospatial attention setting, but XDAWNCov+TS+LDA accuracy was slightly lower here, though not significantly. Therefore, we select the block-Toeplitz linear discriminant analysis decoder for further analyses as the most suitable decoder because of its simplicity.

More interestingly, we noticed that accuracies of the decoders in free visuospatial attention were close to those in the overt visuospatial attention as displayed in fig. 4. A substantial decrease in accuracy from the overt setting to the free setting was observed for subjects C2, C3 and C4 who had the most severe eye motor impairment. For B2, who relied on covert visuospatial attention during the uncued free visuospatial attention according to gaze tracking setting, the decrease in accuracy was also present.

Finally, we report target selection accuracy obtained by averaging multiple trials, a strategy often applied in brain-computer interfaces. This gives a more realistic insight in decoding performance. Full results

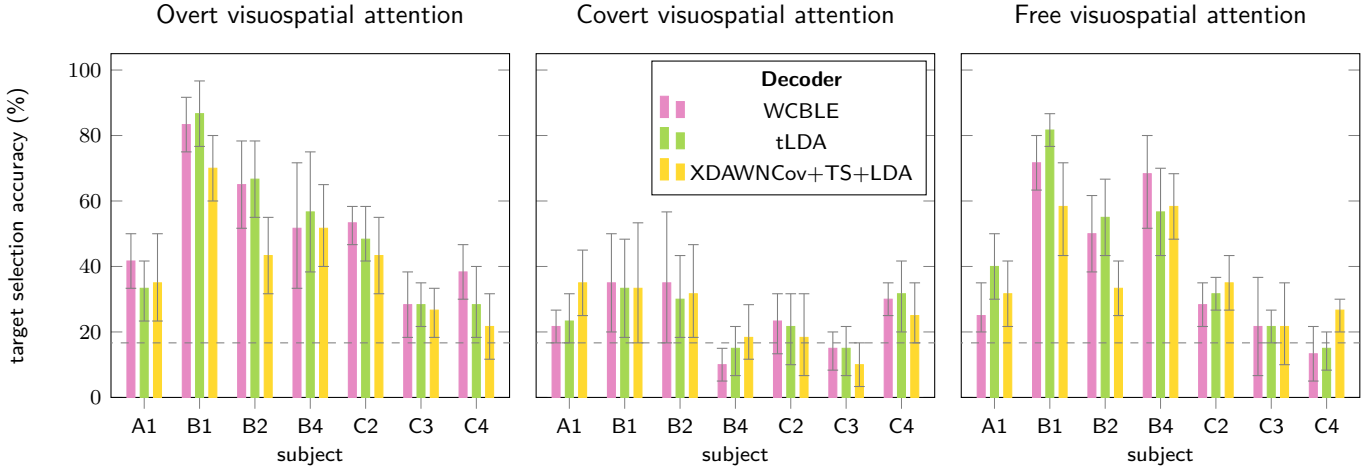


Figure 3: Comparison of single-trial target selection accuracies between the evaluated decoders. Decoders were tested in the three different visuospatial attention settings. Dashed lines indicate the chance level accuracy (1/6). No single decoder yields substantially better performance than the others.

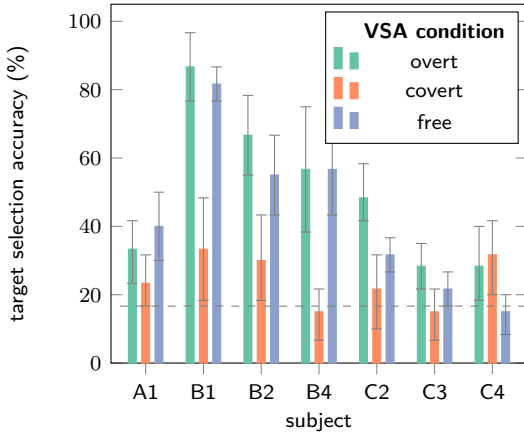


Figure 4: Decoding accuracy compared across visuospatial attention conditions using the block-Toeplitz linear discriminant analysis decoding algorithm. Covert visuospatial attention generally performs poorly, but free visuospatial attention is often on par with overt visuospatial attention, except for participant C2, which performed best with overt attention.

Subject	A1	B1	B2	B4	C2	C3	C4
VSA condition							
covert	-	10	-	-	-	-	9
free	6	1	3	2	9	-	-
overt	9	1	3	2	3	7	-

Table 4: Number of repetitions needed to reach 80% accuracy with the block-Toeplitz linear discriminant analysis decoder

Overt visuospatial attention evaluation Covert visuospatial attention evaluation Free visuospatial attention evaluation

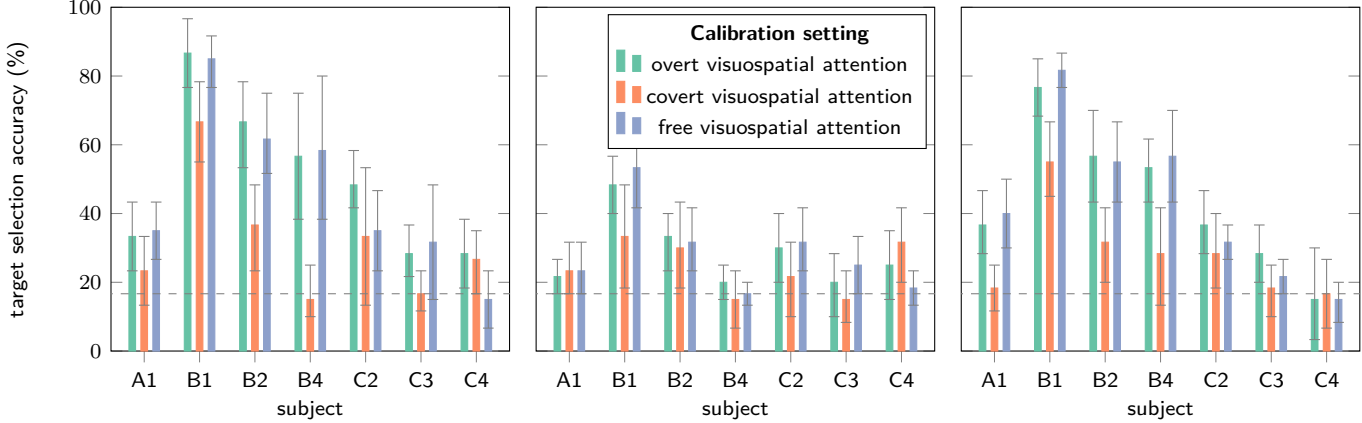


Figure 5: Single-trial target selection accuracy when calibrating the block-Toeplitz linear discriminant analysis decoder in a given visuospatial attention setting, and evaluating it in another. Whiskers indicate confidence intervals. For participants A1, B2, and C3, decoding accuracy in the covert visuospatial attention setting with central gaze fixation improved when calibrating with overt gaze fixation.

are reported in ?? Table 4 reports the number of repetitions needed to reach 80% accuracy, a commonly accepted threshold for effective brain-computer interface decoding [RonAngevin2019] In covert visuospatial attention, only participants B1 and C4 managed to reach acceptable performance, and this with a high amount of repetitions, 10 and 9 respectively. In both free and overt visuospatial attention, most participants managed to achieve acceptable performance after some repetitions.

3.3 Cross-condition decoder training and evaluation

Figure 5 shows that for participants B1 and C2 covert visuospatial attention decoding improved when training with overt visuospatial attention. Note that, according to eye tracking data, participant B1, B4, and C4 did not always perform cued central gaze fixation in the covert visuospatial attention setting, which might have affected the results. A full table of results is available as Additional file 3.